Oleh Basystyi

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Experience

Research and Development Engineer, SoftServe — From Jul 2025

- Designed an indoor surface reconstruction framework for 3DGS.
- Formulated plane assignment as an iterative algorithm, refining both Gaussians and planes.

Research Intern, École de technologie supérieure - ÉTS Montréal — May 2026 - Aug 2026

- Extended **MILo** (Mesh-In-the-Loop Gaussian Splatting), targeting training time, and fine details quality.

Research Intern, Ukrainian Catholic University — Jul 2025 - Jul 2026

- Implemented and compared **filtering algorithms** for IMU-based navigation.
- Evaluated **SLAM** and **VIO** algorithms in low-light conditions, identifying failure modes and research gaps.

Computer Vision Engineer, Alta Ares — Jun 2025 - Sep 2025

- Built a reconstruction pipeline with **COLMAP/GLOMAP** and **Gaussian Splatting (3DGUT)** with object detection using **YOLO**.

Research Intern, UCU Machine Learning Lab — Feb 2025 - Jul 2025

- Investigated efficiency-oriented alternatives to **Gaussian Splatting** and **SfM**.

Teaching Assistant, Ukrainian Catholic University — from Sep 2024

- Assisted in Discrete Mathematics, Computer's Architecture, Algorithms & DSA, Linear Algebra courses;; currently assisting in Artificial Intelligence course.
- Led office hours, mentored for final projects, developed study materials, laboratory works, and homeworks, as well as assessed student progress and graded coursework.

Education

Ukrainian Catholic University, BA Sc. in Computer Science — from Sep 2023

- Ranked among the top 5 Computer Science students in the academic years 2023-2024, 2024-2025 and 2025-2026, with a cumulative GPA 98.2 / 100.

Projects

Simultaneous Uncertainty Low Light Localization And Mapping (SULLLAM) — Repository

- Implemented a Python-based **SLAM** framework integrating uncertainty estimation for low-light environments.

ALEX: A General-Purpose Minecraft AI Agent — Repository

- Designed an open-world AI agent capable of multi-step task planning in Minecraft.

Adaptive Dynamic Hazard Detection (ADHD) — Repository

- Implemented a dynamic change detection pipeline which compared 3D point clouds created with **Fast3R** with detected mapped 3D objects from **YOLO**.

Distributed DL framework — Repository

- Developed a **libtorch**-based framework for distributed deep learning implementing **MPI** and **NCCL** communication primitives with Torch tensors.

MoE-ComputerV — Repository

- Designed and trained a **Mixture-of-Experts** object detection architecture, evaluating custom routing loss functions to optimize expert capacity, routing stability, and task performance against standard baseline models.

Drone Swarm — Repository / Report

- Developed a low-cost computer vision algorithm for LED detection on an embedded device with **OpenCV**, **Eigen** and **Pytorch**. Researched different ways of estimating distance between an FPV drone and object of interest.

Courses & Certifications

AI for Defense: Summer School Jun 2024

– Finished one-week long summer school covering embedded AI applications in military defense units. Implemented a powerline detection algorithm as a final project on the hackathon with **PX4**, **Python**, **YOLO** and **OpenCV**.

"ALHORYTHM" School Sep 2020 - Jan 2023

– Learned the fundamentals of **C++**, **Algorithms** and **Data Structures**. Participated in **contests** and **olympiads** up to national level.

Volunteering

UCU Linux Club from Sep 2025

– Lead, organized and lectured extra classes for first and second year students on Linux.

APPS Summer Camp Jun 2024 - Jul 2024

– Developed a one-week Python course for middle-school students. Held lectures and activities and mentored for final projects.

UCU C++ Club Mar 2024 - May 2025

– Founded the club with the aim to prepare other first-year students for the upcoming courses, held lectures and prepared activities.

Youin Sep 2023 - Jun 2024

– Partook in in the assembly of power banks for Ukraine's Armed Forces. Afterwards, joined the leading coordination team of "Youin" and became part of the R&D team primarily focusing on the construction of drones.